

12.17 (a) First consider longitudinal polarization, $\vec{A} = \vec{E}_3 \cos(kz - \omega t)$. From the Lorenz gauge

condition, $\frac{\partial A^0}{\partial t} + \nabla \cdot \vec{A} = 0$, we have $A^0 = \frac{k}{\omega} \cos(kz - \omega t)$, assuming $c=1$. Then,

$$\vec{E} = -\nabla A^0 - \frac{\partial \vec{A}}{\partial t} = \vec{E}_3 \sin(kz - \omega t) \left(\frac{k^2}{\omega} - \omega \right) = -\vec{E}_3 \frac{\mu^2}{\omega} \sin(kz - \omega t),$$

$$\vec{B} = \nabla \times \vec{A} = 0.$$

Using the result from Problem 12.16(c), we have

$$\langle \mathcal{H}^{00} \rangle = \frac{1}{8\pi} \left(\vec{E}^2 + \vec{B}^2 + \mu^2 (A^0 A^0 + \vec{A} \cdot \vec{A}) \right) = \frac{1}{8\pi} \left(\frac{\mu^4}{\omega^2} \sin^2(kz - \omega t) + \mu^2 \left(\frac{k^2}{\omega^2} + 1 \right) \cos^2(kz - \omega t) \right)$$

$$\langle \mathcal{H}^{30} \rangle = \frac{1}{4\pi} \mu^2 A^3 A^0 = \frac{1}{4\pi} \mu^2 \frac{k}{\omega} \cos^2(kz - \omega t).$$

Time average leads to

$$\langle \mathcal{H}^{00} \rangle = \frac{1}{8\pi} \cdot \frac{1}{2} \left(\frac{\mu^4}{\omega^2} + \frac{\mu^2 k^2}{\omega^2} + \mu^2 \right) = \frac{\mu^2}{8\pi}, \quad \langle \mathcal{H}^{30} \rangle = \frac{\mu^2}{8\pi} \frac{k}{\omega},$$

and $\frac{\langle \mathcal{H}^{30} \rangle}{\langle \mathcal{H}^{00} \rangle} = \frac{k}{\omega}$

For transverse polarization, let us assume $\vec{A} = \vec{E}_1 \cos(kz - \omega t)$. Then, by Lorenz gauge condition,

we can see that $A^0 = 0$. Therefore,

$$\vec{E} = -\frac{\partial \vec{A}}{\partial t} = -\vec{E}_1 \omega \sin(kz - \omega t), \quad \vec{B} = \nabla \times \vec{A} = -\vec{E}_1 k \sin(kz - \omega t).$$

$$\text{Then, } \langle \mathcal{H}^{00} \rangle = \frac{1}{8\pi} \left[\vec{E}^2 + \vec{B}^2 + \mu^2 (A^0 A^0 + \vec{A} \cdot \vec{A}) \right] = \frac{1}{8\pi} \left[(k^2 + \omega^2) \sin^2(kz - \omega t) + \mu^2 \cos^2(kz - \omega t) \right]$$

$$\langle \mathcal{H}^{30} \rangle = \frac{1}{4\pi} \left[(\vec{E} \times \vec{B})_3 + \mu^2 A^3 A^0 \right] = \frac{1}{4\pi} k \omega \sin^2(kz - \omega t).$$

$$\text{and } \langle \mathcal{H}^{00} \rangle = \frac{1}{8\pi} \cdot \frac{1}{2} (k^2 + \omega^2 + \mu^2) = \frac{\omega^2}{8\pi}, \quad \langle \mathcal{H}^{30} \rangle = \frac{k\omega}{8\pi},$$

$$\frac{\langle \mathcal{H}^{30} \rangle}{\langle \mathcal{H}^{00} \rangle} = \frac{k}{\omega}.$$

$$\text{Also, } F_t = \frac{k\omega}{8\pi}, \quad F_L = \frac{\mu^2}{8\pi} \frac{k}{\omega} = \left(\frac{\mu}{\omega} \right)^2 F_t.$$

(b) Details of Thomson scattering can be found in Section 14.8. We are not going to perform strict calculation to obtain the result. Rather, we will gain insights from Thomson scattering of photons. For incident lightwave with $\vec{A} = \vec{E} \cos(kz - \omega t)$, the electric field is given by $\vec{E} = -\frac{\partial \vec{A}}{\partial t} = -\vec{E} \omega \sin(kz - \omega t)$, where the polarization can only be transverse. Then, by Eq. (14.121), the acceleration is

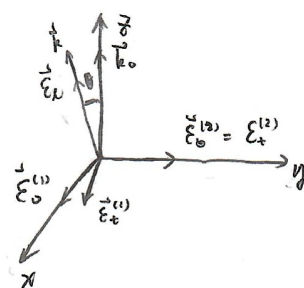
$\vec{\ddot{v}}(t) = -\vec{E} \frac{e}{m} \omega \sin(kz - \omega t)$, and the velocity is $\vec{v}(t) = -\vec{E} \frac{e}{m} \cos(kz - \omega t)$. The outgoing vector potential is then proportional to the velocity, by Eq. (14.8).

Now, for Proca waves with longitudinal polarization, $\vec{E} = -\vec{E} \frac{\mu}{\omega} \sin(kz - \omega t)$. By the same argument as in the photon case, we can see that the outgoing vector potential has an additional factor of $(\mu/\omega)^2$. Also, depending on the polarizations of incident and outgoing waves, the energy fluxes can be different for vector potential of unit strength. Combining all these considerations, we can arrive at

$$\frac{d\sigma}{d\Omega} = r_0^2 E_0 |\vec{\epsilon}^* \cdot \vec{E}_0|^2 \frac{F_{out}}{F_{in}} \quad \text{with } E_0 = \left(\frac{\mu}{\omega}\right)^4 \text{ for longitudinal polarization.}$$

Similarly, for transverse polarization, $\vec{E} = -\vec{E} \omega \sin(kz - \omega t)$, which coincides with the photon case. Therefore, $E_0 = 1$.

(c) For transverse incident wave, assume that $\vec{k}_0$, $\vec{k}$, and $\vec{E}$ are in the same plane, as shown in the figure on the right. Then, $\vec{E}_i$ is in the same plane, while $\vec{E}_r$ is perpendicular. If the outgoing wave is transverse,



$$\vec{E}_0^{(1)} \cdot \vec{E}_t^{(1)} = \cos\theta, \quad \vec{E}_0^{(1)} \cdot \vec{E}_r = \sin\theta, \quad \vec{E}_0^{(1)} \cdot \vec{E}_t^{(2)} = 0$$

$$\vec{E}_0^{(2)} \cdot \vec{E}_t^{(1)} = 0, \quad \vec{E}_0^{(2)} \cdot \vec{E}_r = 0, \quad \vec{E}_0^{(2)} \cdot \vec{E}_t^{(2)} = 1$$

Also, the excitation efficiency $E_0 = 1$ for transverse polarization, and $F_x/F_t = (\mu/\omega)^2$. Then

$$\left(\frac{d\sigma}{d\Omega} \right)_t = r_0^2 \frac{1}{2} \left[1 + \cos^2\theta + \sin^2\theta \frac{F_x}{F_t} \right] = \frac{r_0^2}{2} \left[1 + \cos^2\theta + \left(\frac{\mu}{\omega} \right)^2 \sin^2\theta \right]$$

(d) Similarly,

$$\vec{E}_0^{(1)} \cdot \vec{E}_r = \cos\theta, \quad \vec{E}_0^{(1)} \cdot \vec{E}_t^{(1)} = \sin\theta, \quad \vec{E}_0^{(1)} \cdot \vec{E}_t^{(2)} = 0$$

and $E_0 = (\mu/\omega)^4$ for longitudinal polarization. Then

$$\left(\frac{d\sigma}{d\Omega} \right)_l = r_0^2 \left(\frac{\mu}{\omega} \right)^4 \left[\cos^2\theta + \sin^2\theta \frac{F_t}{F_x} \right] = \left(\frac{\mu}{\omega} \right)^2 r_0^2 \left[\sin^2\theta + \left(\frac{\mu}{\omega} \right)^2 \cos^2\theta \right]$$